Advanced Algorithms

75-Question Multiple-Choice Quiz

Concepts, Proofs, Approximation Ratios, Recurrences, and Randomized Algorithms

Practice Set with Answers and Explanations

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1 Complexity Theory, NP-Completeness, and QSAT

Question 1

Which statement best describes the class **P**?

A. Decision problems solvable in polynomial time by a deterministic algorithm.

B. Decision problems verifiable only in exponential time.

C. Optimization problems that always have an exact linear-time algorithm.

D. Problems requiring nondeterministic exponential time.

Answer: A. Explanation: **P** contains decision problems solvable in time $O(n^c)$ for some constant c on a deterministic machine.

Question 2

Which statement about **NP** is correct?

A. Every problem in NP is known to require exponential time.

B. Every problem in P is also in NP.

C. NP contains only optimization problems.

D. P and NP are known to be unequal.

Answer: B. Explanation: A polynomial-time solution is also a polynomial-time-verifiable certificate, so $\mathbf{P} \subseteq \mathbf{NP}$. Whether $P = NP$ remains open.

Question 3

A decision problem X is NP-complete when which conditions hold?

A. $X \in P$ and every problem in P reduces to X.

B. $X \in NP$ and every problem in NP polynomial-time many-one reduces to X.

C. X has a greedy algorithm and an approximation ratio of 2.

D. X is undecidable and has no finite certificates.

Answer: B. Explanation: NP-completeness requires membership in NP and NP-hardness under polynomial-time reductions.

Question 4

To prove a new decision problem B is NP-hard, which reduction direction is appropriate from a known NP-hard problem A?

A. $B \leq_p A$

B. $A \leq_p B$

C. $A = B$ on all instances

D. B must first be shown to lie in P

Answer: B. Explanation: A polynomial-time reduction $A \leq_p B$ transfers the hardness of A to B. It does not by itself prove $B \in NP$.

Question 5

Which statement about PSPACE is correct?

A. PSPACE consists of problems solvable using polynomial space.

B. PSPACE consists exactly of problems solvable in constant time.

C. PSPACE is known to equal P.

D. PSPACE contains only problems with polynomial-size outputs.

Answer: A. Explanation: PSPACE is defined by polynomially bounded working space; the time used may be much larger.

Question 6

For a fully quantified Boolean formula, which feature makes QSAT a PSPACE-complete problem in its standard form?

A. It has no Boolean variables.

B. It permits alternating existential and universal quantifiers over Boolean variables.

C. It is always solvable by checking one assignment.

D. It is a shortest-path problem.

Answer: B. Explanation: Quantifier alternation represents choices and adversarial responses; deciding truth of quantified Boolean formulas is PSPACE-complete.

Question 7

What does $\exists x\, \forall y\, F(x, y)$ mean?

A. There is one value of x such that $F(x, y)$ is true for every value of y.

B. For every x, there is one value of y making F true.

C. F is true only when $x = y$.

D. Both variables must be assigned randomly.

Answer: A. Explanation: The existential choice of x must work against every possible Boolean value of y.

Question 8

Which statement is generally true about quantifier order?

A. $\exists x \forall y\, F$ is always equivalent to $\forall y \exists x\, F$.

B. Changing quantifier order can change whether a quantified formula is true.

C. Universal and existential quantifiers have identical meanings.

D. Quantifier order matters only for non-Boolean variables.

Answer: B. Explanation: Existential choices may depend on earlier universal choices in one formula but not in the other; the formulas need not be equivalent.

2 Planning and State-Space Search

Question 9

A classical state-space planning problem is commonly specified by which components?

A. A graph, a sorted array, and a pivot.

B. A set of states or conditions, an initial configuration, a goal configuration, and available actions.

C. Only a goal state and a random number.

D. A vertex cover and a matching.

Answer: B. Explanation: Planning models define an initial state, goal condition, and actions/operators with preconditions and effects.

Question 10

In the standard 8-puzzle, what is the role of the blank tile?

A. It is a fixed obstacle that cannot move.

B. It swaps position with a neighboring numbered tile during a legal move.

C. It must always remain in the center.

D. It is counted as a ninth numbered tile.

Answer: B. Explanation: A legal move slides a neighboring tile into the blank, equivalently swapping the blank with that tile.

Question 11

For a permutation puzzle, why can a parity invariant matter when assessing reachability?

A. It proves every arrangement is reachable.

B. It can show that certain configurations cannot be reached from the initial configuration by legal moves.

C. It determines the weight of each tile.

D. It guarantees a greedy algorithm is optimal.

Answer: B. Explanation: Legal moves preserve particular parity properties, so a goal with incompatible parity may be unreachable.

Question 12

If a search tree has branching factor b and depth d, what is the usual upper bound on the number of nodes through depth d for $b > 1$?

A. $O(d + b)$

B. $O(b^d)$

C. $O(d^b)$

D. $O(\log_b d)$

Answer: B. Explanation: The geometric sum $1 + b + \cdots + b^d$ is $O(b^d)$ for fixed $b > 1$.

Question 13

Which distinction is most accurate?

A. A planning problem asks for a sequence of actions reaching a goal; an optimization version may additionally minimize a cost such as plan length.

B. Planning never uses states.

C. Every planning problem is automatically NP-complete.

D. Optimization problems have no objective function.

Answer: A. Explanation: Feasibility asks whether a goal can be reached; optimization asks for a best feasible plan under a specified cost.

3 Vertex Cover and Pricing-Based Approximation

Question 14

A set $C \subseteq V$ is a vertex cover of an undirected graph $G = (V, E)$ when:

A. Every vertex has degree at least one.

B. Every edge has at least one endpoint in C.

C. Every edge has both endpoints in C.

D. No two vertices in C are adjacent.

Answer: B. Explanation: The defining condition is that for every edge (u, v), $u \in C$ or $v \in C$ (or both).

Question 15

The standard edge-picking 2-approximation repeatedly chooses an uncovered edge and adds both endpoints. Why is its output a vertex cover?

A. Every edge is eventually deleted only if its endpoints are both outside the cover.

B. Each chosen edge's endpoints are added, and all edges incident to selected vertices are thereby covered.

C. It selects exactly one endpoint of every edge.

D. It requires the graph to be bipartite.

Answer: B. Explanation: When the algorithm stops, no uncovered edge remains; each removed edge is incident to a selected endpoint.

Question 16

In the edge-picking Vertex Cover algorithm, selected edges are chosen so they are pairwise vertex-disjoint. What follows for an optimal cover C^*?

A. $|C^*|$ is at least the number of selected edges.

B. $|C^*|$ is at most the number of selected edges.

C. $|C^*| = 2|A|$ always.

D. The selected edges form a Hamiltonian cycle.

Answer: A. Explanation: A cover must include at least one endpoint of each edge in a matching, and disjoint edges require distinct cover vertices.

Question 17

If the algorithm selects q pairwise vertex-disjoint edges and returns both endpoints of each, what bound follows?

A. $|C| = q$ and $|C| \leq |C^*|$

B. $|C| = 2q$ and $|C^*| \geq q$, so $|C| \leq 2|C^*|$

C. $|C| = q/2$ and $|C^*| \leq q$

D. $|C| = 2|C^*|$ exactly

Answer: B. Explanation: The returned cover contains $2q$ endpoints, while every cover has size at least q.

Question 18

In Weighted Vertex Cover, each vertex v has weight $w_v \geq 0$. What is the objective?

A. Minimize the number of edges.

B. Minimize $\sum_{v \in C} w_v$ subject to every edge being covered.

C. Maximize the number of selected vertices.

D. Minimize the maximum degree only.

Answer: B. Explanation: The cover must be feasible and have minimum total vertex weight.

Question 19

Which constraints make edge prices p_e feasible for the pricing method for Weighted Vertex Cover?

A. $p_e \leq 0$ and $\sum_{e \ni v} p_e \geq w_v$

B. $p_e \geq 0$ and $\sum_{e \ni v} p_e \leq w_v$ for every vertex v

C. $p_e = w_v$ for every incident edge

D. $\sum_e p_e = 0$

Answer: B. Explanation: Prices are nonnegative and the sum of prices on edges incident to each vertex cannot exceed its weight.

Question 20

Given feasible edge prices, which inequality is guaranteed for every vertex cover S?

A. $w(S) \leq \sum_e p_e$

B. $w(S) = \sum_e p_e$

C. $\sum_e p_e \leq w(S)$

D. $2w(S) \leq \sum_e p_e$

Answer: C. Explanation: Every edge has at least one endpoint in S, so its price is counted at least once in the incident-price sum over S; fairness bounds that sum by $w(S)$.

Question 21

A vertex v is tight in the pricing method when:

A. $w_v = 0$ only.

B. $\sum_{e \ni v} p_e = w_v$.

C. v has maximum degree.

D. Every edge in the graph has price zero.

Answer: B. Explanation: Tightness means the incident edge prices exactly exhaust the vertex's weight budget.

Question 22

If an algorithm returns only tight vertices and each edge price is counted at most twice in the selected-vertex sum, which guarantee follows?

A. $w(S) \leq \frac{1}{2}\mathrm{OPT}$

B. $w(S) \leq 2\mathrm{OPT}$

C. $w(S) = \mathrm{OPT}$ always

D. $w(S) \geq 4\mathrm{OPT}$

Answer: B. Explanation: Tightness gives $w(S) = \sum_{v \in S} \sum_{e \ni v} p_e \leq 2 \sum_e p_e \leq 2\mathrm{OPT}$.

Question 23

Which theorem relates minimum Vertex Cover and maximum Matching in a bipartite graph?

A. Euler's theorem

B. König's theorem

C. Bayes' theorem

D. Fermat's little theorem

Answer: B. Explanation: König's theorem states that the minimum vertex-cover size equals the maximum matching size in every bipartite graph.

4 Metric Traveling Salesperson Problem

Question 24

Which property is required for the standard MST-based 2-approximation for metric TSP?

A. Triangle inequality for distances or edge costs.

B. All edges have identical weights.

C. The graph must be a tree.

D. The optimal tour must be known beforehand.

Answer: A. Explanation: Shortcutting repeated vertices relies on the triangle inequality not increasing tour cost.

Question 25

Why is the weight of an MST T no greater than the weight of an optimal Hamiltonian cycle H^*?

A. Every MST contains all edges of H^*.

B. Deleting one edge from H^* leaves a spanning tree, and the MST is no more expensive than that tree.

C. The triangle inequality makes every MST a Hamiltonian cycle.

D. An MST always has exactly the same edges as the optimal tour.

Answer: B. Explanation: Removing one tour edge produces a spanning tree of cost no greater than the tour; the MST is the minimum-cost spanning tree.

Question 26

If $w(T) = 25$, what upper bound does the standard MST-preorder algorithm guarantee for its metric-TSP tour cost?

A. 25

B. 37.5

C. 50

D. 625

Answer: C. Explanation: The preorder shortcut tour has cost at most $2w(T) = 50$. This is an upper bound, not necessarily the actual tour cost.

Question 27

What does the triangle inequality guarantee when a repeated vertex in a closed tree walk is shortcut?

A. The tour cost cannot increase.

B. The tour cost must decrease strictly.

C. The shortcut edge must belong to the MST.

D. The resulting route is always optimal.

Answer: A. Explanation: A direct edge costs no more than the path through the skipped vertices, so shortcutting does not increase cost.

Question 28

Which is the standard approximation guarantee for the MST-preorder algorithm on metric TSP?

A. 1-approximation

B. 2-approximation

C. $\log n$-approximation only

D. No finite guarantee even for metric costs

Answer: B. Explanation: The tour cost is at most $2w(T)$, and $w(T) \leq w(H^*)$, giving $w(H) \leq 2w(H^*)$.

Question 29

For a complete graph with n vertices, Kruskal's algorithm typically gives which asymptotic bound for constructing the MST?

A. $O(n)$

B. $O(n \log n)$

C. $O(n^2 \log n)$

D. $O(2^n)$

Answer: C. Explanation: A complete graph has $m = \Theta(n^2)$ edges, and Kruskal takes $O(m \log m) = O(n^2 \log n)$.

5 Load Balancing and Scheduling

Question 30

For jobs with processing times t_j assigned to machines M_i, the load of machine i is:

A. $T_i = \max_{j \in A(i)} t_j$

B. $T_i = \sum_{j \in A(i)} t_j$

C. $T_i = \sum_i t_i$

D. $T_i = |A(i)|/m$

Answer: B. Explanation: Machine load is the sum of processing times of all jobs assigned to that machine.

Question 31

What objective does identical-machine makespan scheduling minimize?

A. The total sum of all processing times, which is fixed.

B. The minimum load of any machine.

C. The maximum machine load $\max_i T_i$.

D. The number of machines, even when fixed.

Answer: C. Explanation: Makespan is the largest machine load, and scheduling seeks to minimize it.

Question 32

Which two lower bounds hold for optimal makespan T^* on m identical machines?

A. $T^* \leq \sum_j t_j/m$ and $T^* \leq \max_j t_j$

B. $T^* \geq \sum_j t_j/m$ and $T^* \geq \max_j t_j$

C. $T^* = \sum_j t_j/m$ always

D. $T^* \geq m \sum_j t_j$

Answer: B. Explanation: Some machine must carry at least the average load, and the largest job must fit on some machine.

Question 33

What does List Scheduling do when the next job arrives?

A. Assign it to a currently least-loaded machine.

B. Assign it to the currently most-loaded machine.

C. Sort all jobs after every assignment.

D. Reject the job if its duration exceeds the average.

Answer: A. Explanation: The greedy rule assigns each next job to a machine with minimum current load.

Question 34

What is the standard worst-case approximation bound for List Scheduling on m identical machines?

A. $T \leq (2 - 1/m)T^*$

B. $T \leq (1 + 1/m)T^*$

C. $T \leq T^*/m$

D. $T \leq m^2 T^*$

Answer: A. Explanation: The standard bound is $2 - 1/m$, derived by examining the last-finishing job and applying the average-load lower bound.

Question 35

For List Scheduling with $m = 2$, the standard guarantee becomes:

A. $T \leq T^*$

B. $T \leq \frac{3}{2}T^*$

C. $T \leq 2T^*$ exactly

D. $T \leq 3T^*$

Answer: B. Explanation: Substitute $m = 2$ into $2 - 1/m$ to obtain $3/2$.

Question 36

How does Longest Processing Time first (LPT) differ from ordinary List Scheduling?

A. LPT sorts jobs in nonincreasing processing-time order before assigning each to a least-loaded machine.

B. LPT always assigns the shortest job to the most-loaded machine.

C. LPT uses no machines' current loads.

D. LPT solves every scheduling instance optimally.

Answer: A. Explanation: LPT adds a decreasing-time ordering before applying the least-loaded-machine rule.

Question 37

Which is the classical LPT approximation bound for m identical machines?

A. $T \leq (4/3 - 1/(3m))T^*$

B. $T \leq (2m)T^*$

C. $T \leq T^*/2$

D. $T \leq \log(m)T^*$

Answer: A. Explanation: The classical LPT bound is $4/3 - 1/(3m)$; it is stronger than the general List Scheduling bound.

6 Metric k-Center

Question 38

What objective does metric k-Center minimize?

A. The sum of all pairwise distances among sites.

B. The maximum, over all sites, of the distance to the nearest selected center.

C. The number of edges in the input graph.

D. The maximum distance between any two selected centers.

Answer: B. Explanation: The covering radius is $r(C) = \max_{s \in S} \min_{c \in C} d(s, c)$.

Question 39

Which is the triangle inequality for a metric distance d?

A. $d(a, b) \geq d(a, c) + d(c, b)$

B. $d(a, b) \leq d(a, c) + d(c, b)$

C. $d(a, b) = 0$ for all a, b

D. $d(a, b) = -d(b, a)$

Answer: B. Explanation: A direct distance is no greater than the length of a route through an intermediate point.

Question 40

After selecting some centers C, which site does farthest-first traversal select next?

A. The site minimizing $d(s, C)$.

B. The site maximizing $d(s, C) = \min_{c \in C} d(s, c)$.

C. The site with maximum degree in every metric instance.

D. A site chosen outside the input set.

Answer: B. Explanation: It chooses the currently worst-served site, i.e. the one farthest from its nearest selected center.

Question 41

What is the standard approximation guarantee for farthest-first traversal in metric k-Center?

A. $r(C) \leq r^*/2$

B. $r(C) \leq 2r^*$

C. $r(C) = r^*$ for every input

D. $r(C) \leq nr^*$ only

Answer: B. Explanation: The packing argument and triangle inequality prove a factor-two covering-radius guarantee.

Question 42

In the metric k-Center 2-approximation proof, why are $k + 1$ separated sites compared with k optimal centers?

A. To force two of the sites to be assigned to the same optimal center.

B. To prove the graph is complete.

C. To show every center has degree k.

D. To make the optimal radius equal to zero.

Answer: A. Explanation: With more sites than optimal centers, two sites must share an optimal center; the triangle inequality bounds their distance by $2r^*$.

Question 43

A greedy radius procedure selects a site and removes all remaining sites within distance $2r$. If it selects more than k centers, what can be concluded?

A. A radius-r cover by at most k centers exists.

B. No cover of radius r using at most k centers exists.

C. The optimal radius is exactly $2r$.

D. The graph violates symmetry.

Answer: B. Explanation: Selected sites are pairwise more than $2r$ apart, so each would need a distinct optimal radius-r center.

Question 44

With incremental nearest-center distances, what is the typical running time of farthest-first traversal on n sites selecting k centers?

A. $O(nk)$

B. $O(1)$

C. $O(2^n)$ necessarily

D. $O(k/n)$

Answer: A. Explanation: Each of the k rounds scans or updates distances for n sites in constant time per site.

7 Set Cover

Question 45

Given universe U and subsets $S_1, \ldots, S_m$, a valid cover must satisfy:

A. $\bigcup_{S_i \in \mathcal{C}} S_i = U$

B. $\bigcap_{S_i \in \mathcal{C}} S_i = U$

C. $|S_i| = 1$ for every chosen set

D. $\mathcal{C}$ must contain all subsets

Answer: A. Explanation: Every universe element must belong to at least one selected subset.

Question 46

For weighted Set Cover, what is the objective?

A. Minimize the sum of costs of selected subsets while covering all elements.

B. Minimize the number of elements in the universe.

C. Maximize the cost of the selected subsets.

D. Choose exactly one subset regardless of coverage.

Answer: A. Explanation: The cover must cover the universe and have minimum total set cost.

Question 47

In weighted greedy Set Cover, which subset is selected at each step?

A. The subset maximizing cost divided by newly covered elements.

B. The subset minimizing cost per newly covered element.

C. The subset with the largest original size regardless of cost or coverage.

D. The subset with the fewest uncovered elements.

Answer: B. Explanation: Choose a set minimizing $w_i/|S_i \cap R|$, where R is the currently uncovered universe.

Question 48

Why must the greedy Set Cover algorithm recalculate newly covered elements after each selection?

A. Previously covered elements no longer contribute to marginal coverage.

B. Set costs become negative automatically.

C. The universe changes its labels.

D. The optimal solution is then known exactly.

Answer: A. Explanation: The relevant quantity is $|S_i \cap R|$; elements already covered are not new contributions.

Question 49

What is the standard approximation guarantee for greedy weighted Set Cover on a universe of n elements?

A. H_n-approximation, where $H_n = 1 + 1/2 + \cdots + 1/n$

B. $1/n$-approximation

C. 2-approximation for all instances

D. Exact optimality for all instances

Answer: A. Explanation: The greedy solution has cost at most H_n times optimum, and $H_n \leq 1+\ln n$.

Question 50

Which inequality relates the harmonic number to a logarithm?

A. $H_n \leq 1 + \ln n$

B. $H_n \geq n^2$

C. $H_n = \ln(1/n)$

D. $H_n \leq 1/n$

Answer: A. Explanation: The harmonic sum grows logarithmically and satisfies $H_n \leq 1 + \ln n$.

Question 51

If every set has the same cost, the weighted greedy criterion reduces to choosing the set that:

A. Covers the largest number of currently uncovered elements.

B. Contains the largest number of already-covered elements.

C. Has the smallest original cardinality.

D. Contains no new elements.

Answer: A. Explanation: With equal costs, minimizing cost per newly covered element is equivalent to maximizing newly covered elements.

8 Local Search and Bipartite Vertex Cover

Question 52

A local optimum with respect to a neighborhood is a solution that:

A. Is globally best among every feasible solution.

B. Has no better solution among its permitted neighboring solutions.

C. Must be found in one iteration.

D. Always has an approximation ratio of one.

Answer: B. Explanation: Local optimality depends on the chosen neighborhood and does not imply global optimality.

Question 53

Which statement correctly distinguishes a global optimum from a local optimum?

A. A global optimum is best over all feasible solutions; a local optimum is best only within its neighborhood.

B. They are synonyms for every combinatorial problem.

C. A local optimum always has lower cost than a global optimum.

D. A global optimum is defined only for continuous functions.

Answer: A. Explanation: The global optimum compares against the entire feasible space; local optimality compares only against allowed nearby changes.

Question 54

What defines a bipartite graph with partitions L and R?

A. Every edge joins two vertices in L.

B. Every edge joins a vertex in L to one in R.

C. Every vertex has degree two.

D. The graph must be complete.

Answer: B. Explanation: The vertex set is split into two parts and no edge lies within a single part.

Question 55

For complete bipartite graph $K_{2,3}$, what is the minimum vertex-cover size?

A. 1

B. 2

C. 3

D. 5

Answer: B. Explanation: Selecting the two vertices on the smaller side covers all edges; every cover must include one whole side in this complete bipartite graph.

Question 56

Which equality holds for every bipartite graph?

A. Minimum Vertex Cover size equals maximum Matching size.

B. Minimum Vertex Cover size equals number of connected components.

C. Maximum Matching size equals number of vertices.

D. Minimum Vertex Cover weight equals maximum edge weight.

Answer: A. Explanation: König's theorem equates maximum matching cardinality and minimum vertex-cover cardinality in bipartite graphs.

9 Metropolis Algorithm and Hopfield Networks

Question 57

For an energy change $\Delta E = E(x') - E(x) > 0$, what is the Metropolis acceptance probability at temperature $T > 0$?

A. $e^{\Delta E/T}$

B. $e^{-\Delta E/T}$

C. $1 - \Delta ET$

D. $T/\Delta E$

Answer: B. Explanation: An uphill move is accepted with probability $e^{-\Delta E/T}$, which lies between zero and one.

Question 58

If a proposed move has $\Delta E < 0$, what does the standard Metropolis rule do?

A. Accept it with probability one.

B. Reject it with probability one.

C. Accept it only when $T = 0$.

D. Choose a new initial state.

Answer: A. Explanation: A lower-energy move is accepted; the probabilistic rule is needed for uphill moves.

Question 59

For a fixed positive energy increase ΔE, what happens to $e^{-\Delta E/T}$ as T increases?

A. It decreases toward zero.

B. It increases toward one.

C. It remains exactly constant.

D. It becomes negative.

Answer: B. Explanation: A higher temperature makes uphill moves more likely because the negative exponent has smaller magnitude.

Question 60

Which is a geometric cooling schedule?

A. $T_{t+1} = T_t + 1$

B. $T_{t+1} = \alpha T_t$ with $0 < \alpha < 1$

C. $T_{t+1} = T_t^2 + 1$

D. $T_{t+1} = 0$ for every t

Answer: B. Explanation: Multiplication by a constant factor between zero and one decreases temperature geometrically.

Question 61

Why can simulated annealing escape a local minimum more readily than strict greedy descent?

A. It sometimes accepts moves that increase the objective, with probability controlled by temperature.

B. It never evaluates objective changes.

C. It sorts all states exactly.

D. It requires the global optimum as its initial state.

Answer: A. Explanation: Accepting some uphill moves allows the search to leave a local minimum; cooling gradually reduces this behavior.

Question 62

In the standard binary Hopfield model, a neuron's state is commonly represented by:

A. $s_i \in \{-1, +1\}$

B. $s_i \in \{0, 100\}$ only

C. s_i must be a permutation

D. s_i must be a positive edge weight

Answer: A. Explanation: The common bipolar encoding uses -1 and +1 for the two neuron states.

Question 63

Which condition is normally used for the standard Hopfield energy argument?

A. $w_{ij} = w_{ji}$ and $w_{ii} = 0$

B. $w_{ij} = -w_{ji}$ for all pairs

C. All weights must be positive

D. Every neuron must connect to every other neuron

Answer: A. Explanation: Symmetric weights and zero self-connections support monotonic energy descent under asynchronous updates.

Question 64

For $h_i = \sum_j w_{ij} s_j$ and energy $E = \sum_{(i,j)} w_{ij} s_i s_j$, what is the energy change when only s_i flips?

A. $\Delta E = 2 s_i h_i$

B. $\Delta E = -2 s_i h_i$

C. $\Delta E = h_i^2$

D. $\Delta E = 0$ always

Answer: B. Explanation: The terms incident to i change from $s_i h_i$ to $-s_i h_i$, so the difference is $-2 s_i h_i$.

10 Randomized Quickselect and Quicksort

Question 65

For the sorted values $10, 15, 42, 55, 68, 72, 77$, what is the fourth-smallest element?

A. 42

B. 55

C. 68

D. 72

Answer: B. Explanation: With seven sorted elements, the fourth element is the median, 55.

Question 66

After partitioning around pivot p, let ℓ elements be smaller than p. What is the pivot's rank in the current subarray, using one-based ranks?

A. ℓ

B. $\ell + 1$

C. $n - \ell$

D. $n + 1$

Answer: B. Explanation: Exactly ℓ elements precede the pivot, so its rank is $\ell + 1$.

Question 67

If the desired rank is $k > \ell + 1$, what rank should be searched for in the right partition?

A. k

B. $k - \ell$

C. $k - \ell - 1$

D. $\ell - k + 1$

Answer: C. Explanation: The right partition excludes the ℓ smaller elements and the pivot, so subtract $\ell + 1$ from k.

Question 68

If each pivot splits the current array exactly in half, which recurrence and bound describe Quickselect?

A. $T(n) = 2T(n/2) + cn = \Theta(n \log n)$

B. $T(n) = T(n/2) + cn = \Theta(n)$

C. $T(n) = T(n-1) + cn = \Theta(\log n)$

D. $T(n) = T(n/4) + c = \Theta(1)$

Answer: B. Explanation: Quickselect recurses into only one half; the linear work forms a geometric series.

Question 69

If Quickselect repeatedly chooses an extreme pivot, what recurrence and worst-case bound result?

A. $T(n) = T(n-1) + cn = \Theta(n^2)$

B. $T(n) = T(n/2) + cn = \Theta(n)$

C. $T(n) = 2T(n/2) + cn = \Theta(n)$

D. $T(n) = T(1) + c = \Theta(1)$

Answer: A. Explanation: Partition sizes decrease by only one, so the total work is $c(n + (n-1) + \cdots + 1) = \Theta(n^2)$.

Question 70

What is the expected running time of randomized Quickselect with a uniformly random pivot?

A. $\Theta(1)$

B. $\Theta(\log n)$

C. $\Theta(n)$

D. $\Theta(n \log n)$

Answer: C. Explanation: The expected total partitioning work is linear, although the worst case remains quadratic.

Question 71

If a pivot is called good when both resulting sides have size at most about $3n/4$, what lower bound holds for its probability under a uniformly random pivot, ignoring rounding details?

A. $1/8$

B. $1/4$

C. $1/2$

D. 1

Answer: C. Explanation: At least the middle half of the sorted ranks produce a split with neither side exceeding about $3n/4$.

Question 72

For a balanced Quicksort split, which recurrence applies?

A. $T(n) = T(n/2) + cn$

B. $T(n) = 2T(n/2) + cn$

C. $T(n) = T(n-1) + c$

D. $T(n) = T(n/2) + c$

Answer: B. Explanation: Quicksort recursively sorts both halves and performs linear partitioning work.

Question 73

What is the expected running time of randomized Quicksort on distinct keys?

A. $\Theta(1)$

B. $\Theta(n)$

C. $\Theta(n \log n)$

D. $\Theta(n^2 \log n)$

Answer: C. Explanation: Random pivots yield expected logarithmic recursion depth and linear work per level.

Question 74

Which statement correctly compares Quickselect and Quicksort?

A. Quickselect recursively processes one relevant partition; Quicksort processes both partitions.

B. Both always recurse into both partitions.

C. Quickselect sorts the entire array, while Quicksort finds only the median.

D. Both have guaranteed worst-case $\Theta(n \log n)$ time with ordinary random pivots.

Answer: A. Explanation: Selection needs only the partition containing the requested rank, while sorting must sort both sides.

Question 75

Which of the following statements are correct? *(Select all that apply.)*

A. Randomized Quickselect can have expected $\Theta(n)$ time.

B. Randomized Quicksort has expected $\Theta(n \log n)$ time.

C. Both algorithms have worst-case $\Theta(n^2)$ time with ordinary pivot partitioning.

D. Randomization guarantees every individual run is balanced.

Answer: A, B, C. Explanation: Quickselect recurses into one side, Quicksort recurses into both, and extreme pivots can still cause quadratic worst cases; randomization improves expectation, not every run.

Quick Formula Sheet

Quick Formula Sheet

Algorithm / Concept	Formula / Guarantee
Vertex Cover (edge-picking)	$\lvert C\rvert \le 2\lvert C^*\rvert$
Metric TSP (MST preorder)	$w(H) \le 2w(H^*)$
List Scheduling	$T \le \left(2 - \frac{1}{m}\right) T^*$
LPT Scheduling	$T \le \left(\frac{4}{3} - \frac{1}{3m}\right) T^*$
Metric k-Center (farthest-first)	$r(C) \le 2r^*$
Greedy Set Cover	$C_{\text{greedy}} \le H_n C^* \le (1 + \ln n) C^*$
Pricing lower bound	$\sum_{e \in E} p_e \le \text{OPT}$
Metropolis (uphill acceptance)	$p = e^{-\Delta E/T}, \quad \Delta E > 0$
Quickselect (expected time)	$\Theta(n)$
Randomized Quicksort (expected time)	$\Theta(n \log n)$
Quickselect / Quicksort (worst case)	$\Theta(n^2)$

Advanced Algorithms

Challenge Quiz, Part II: Questions 76–150

Tricky MCQs, Multi-Select, Proofs, and Approximation Analysis

Continuation of the 75-Question Practice Set

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1 Complexity and Reductions **2**

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1 Complexity and Reductions

Question 76

Suppose A reduces to B in polynomial time and B is in P. What follows?

A. A is in P

B. B is NP-complete

C. A is NP-hard

D. None of these

Answer: A. Explanation: Run the reduction, then solve B; both steps take polynomial time.

Question 77

Which proves B is NP-complete?

A. B is in NP and B reduces to known NP-complete A

B. B is in NP and known NP-complete A reduces to B

C. B is NP-hard only

D. One instance of B takes exponential time

Answer: B. Explanation: Membership in NP plus a reduction from a known NP-complete problem proves NP-completeness.

Question 78

Which statements about polynomial-time reductions are correct? *(Select all that apply.)*

A. If A reduces to B and B is in P, then A is in P

B. If A is NP-hard and A reduces to B, then B is NP-hard

C. A reduction is automatically reversible

D. A many-one reduction preserves yes/no answers

Answer: A, B, D. Explanation: Reductions are directional and preserve the decision answer.

Question 79

Which inclusion is known unconditionally?

A. P equals NP

B. NP is a subset of PSPACE

C. PSPACE is a subset of P

D. None of these

Answer: B. Explanation: The standard inclusions are P subseteq NP subseteq PSPACE.

Question 80

What does exists x for all y F(x,y) mean?

A. For every x some y works

B. One fixed x works for every y

C. Some pair x,y works

D. None of these

Answer: B. Explanation: The existential choice must satisfy F for every Boolean value of y.

Question 81

Let F(x,y) be x iff y. Is exists x for all y F(x,y) true?

A. Yes, choose x=0

B. Yes, choose x=1

C. No, no fixed x equals both values of y

D. None of these

Answer: C. Explanation: For any fixed x, the equivalence fails for the opposite value of y.

Question 82

Which QBF statements are correct? *(Select all that apply.)*

A. Quantifier order can affect truth

B. Universal quantification checks every Boolean value

C. Existential quantification checks every Boolean value

D. TQBF is decidable in polynomial space

Answer: A, B, D. Explanation: Existential quantification needs at least one satisfying value; TQBF is PSPACE-complete.

2 Planning and Search

Question 83

In classical planning, when is an action applicable?

A. When its effects equal the goal

B. When its preconditions hold in the current state

C. When it reduces the variable count

D. None of these

Answer: B. Explanation: Preconditions determine applicability; effects define the successor state.

Question 84

For fixed branching factor b greater than 1, nodes through depth d total:

A. $\Theta(b^d)$

B. $\Theta(d^b)$

C. Theta(b+d)

D. None of these

Answer: A. Explanation: The geometric sum $1 + b + \cdots + b^d$ is $\Theta(b^d)$.

Question 85

How many arrangements of eight numbered tiles plus a blank exist before reachability restrictions?

A. 8!

B. 9!

C. 9 squared

D. None of these

Answer: B. Explanation: There are nine distinct objects, so there are 9! permutations.

Question 86

A legal 8-puzzle move:

A. Moves any tile to any empty location

B. Swaps the blank with an orthogonally adjacent tile

C. Swaps any two numbered tiles

D. None of these

Answer: B. Explanation: Only a neighboring tile can slide into the blank.

Question 87

Which statements about state-space search are correct? *(Select all that apply.)*

A. A state graph may contain cycles

B. A plan reaches a goal from the initial state

C. Every reachable goal has a unique shortest plan

D. Repeated-state checking can avoid redundant exploration

Answer: A, B, D. Explanation: Multiple shortest plans can exist; duplicate detection can prevent repeated work.

3 Vertex Cover and Pricing

Question 88

For edges (a,b),(b,c),(c,d), which is a vertex cover of size one?

A. b

B. c

C. a

D. None of these

Answer: D. Explanation: No single vertex touches all three edges.

Question 89

The complement of a vertex cover is always:

A. A clique

B. An independent set

C. A matching

D. A Hamiltonian cycle

Answer: B. Explanation: Two adjacent vertices outside the cover would leave their edge uncovered.

Question 90

If M is a matching, what lower bound does it give on minimum vertex-cover size tau?

A. tau <= |M|/2

B. tau >= |M|

C. tau = 2|M|

D. None of these

Answer: B. Explanation: Each disjoint matching edge needs a distinct selected endpoint.

Question 91

Which claims about the edge-picking 2-approximation are correct? *(Select all that apply.)*

A. It adds both endpoints of an uncovered edge

B. Selected edges form a matching after incident edges are removed

C. Output size is at most twice optimum

D. It always finds an optimum cover

Answer: A, B, C. Explanation: If q disjoint edges are picked, output size is 2q while OPT is at least q.

Question 92

The algorithm selects seven pairwise disjoint edges and returns both endpoints. What follows?

A. It returns 7 vertices and OPT >= 14

B. It returns 14 vertices and OPT >= 7

C. It returns 14 vertices and OPT <= 7

D. None of these

Answer: B. Explanation: The algorithm returns 2q vertices and any cover needs at least q.

Question 93

For a minimization problem, a rho-approximation guarantees:

A. OPT/ALG <= rho

B. ALG/OPT <= rho

C. ALG equals rho times OPT

D. None of these

Answer: B. Explanation: The ratio is ALG/OPT, and the output cost is at most rho times optimum.

Question 94

Feasible edge prices for weighted Vertex Cover satisfy:

A. Prices are negative and incident sum >= weight

B. Prices are nonnegative and incident sum <= vertex weight

C. Every edge price equals both endpoint weights

D. None of these

Answer: B. Explanation: The dual constraints are $p_e >= 0$ and sum of incident prices <= w_v.

Question 95

If feasible edge prices total 18, what follows for optimum weighted cover cost?

A. OPT <= 9

B. OPT = 18

C. OPT >= 18

D. No bound follows

Answer: C. Explanation: Every cover costs at least the total feasible edge price; the bound need not be tight.

Question 96

S contains only tight vertices and each edge price is counted at most twice. Which follow? *(Select all that apply.)*

A. Weight of S equals its incident-price sum

B. Weight of S <= twice total edge price

C. Feasible total edge price <= OPT

D. Weight of S <= OPT/2 necessarily

Answer: A, B, C. Explanation: Tightness gives equality, double counting gives factor two, and feasible prices lower-bound OPT.

4 Metric TSP and MST

Question 97

If an MST has weight 31, the MST-preorder metric-TSP tour is bounded above by:

A. 15.5

B. 31

C. 62

D. None of these

Answer: C. Explanation: The standard tour cost is at most twice the MST weight.

Question 98

Why is triangle inequality needed for shortcutting repeated vertices?

A. To prove the MST unique

B. To shortcut without increasing cost

C. To make all edges equal

D. None of these

Answer: B. Explanation: A direct edge costs no more than the path through skipped vertices.

Question 99

Which lower bound holds for metric TSP optimum OPT and MST T?

A. OPT <= w(T)

B. w(T) <= OPT

C. 2OPT <= w(T) always

D. None of these

Answer: B. Explanation: Deleting a tour edge leaves a spanning tree no more expensive than the tour; the MST is no heavier.

Question 100

A tour costs 84 and a valid lower bound says OPT >= 42. Which is justified?

A. Tour is optimal

B. Tour is within factor 2 of optimum

C. OPT must equal 84

D. None of these

Answer: B. Explanation: ALG/OPT <= 84/42 = 2.

Question 101

Which are true of the MST-based metric-TSP algorithm? *(Select all that apply.)*

A. MST weight lower-bounds optimal tour

B. Preorder traversal first visits each tree vertex once on entry

C. Triangle-inequality shortcutting can increase cost

D. Guarantee is at most factor 2

Answer: A, B, D. Explanation: Shortcutting cannot increase cost, so tour <= 2 MST <= 2 OPT.

Question 102

A complete graph on n vertices has how many undirected edges?

A. n-1

B. n

C. n(n-1)/2

D. 2^n

Answer: C. Explanation: Every unordered pair of distinct vertices forms one edge.

5 Load Balancing

Question 103

Machine loads are 8, 11, and 6. The makespan is:

A. 6

B. 8

C. 11

D. 25

Answer: C. Explanation: Makespan is the maximum machine load.

Question 104

Jobs have times 9,7,6,4 on two machines. Strongest listed lower bound on OPT?

A. Average bound only 13

B. Largest-job bound only 9

C. Maximum of bounds 13

D. None of these

Answer: C. Explanation: Total work is 26, so average load is 13, which exceeds the largest job 9.

Question 105

The last job starts on a least-loaded machine with prior load L. Which holds?

A. L <= total processing time divided by m

B. L is always zero

C. L >= OPT

D. None of these

Answer: A. Explanation: The chosen machine is least loaded, so its prior load cannot exceed the average load so far.

Question 106

The standard List Scheduling proof gives:

A. T <= (2-1/m)OPT

B. T <= OPT/m

C. T <= m OPT squared

D. None of these

Answer: A. Explanation: For the last job j, $T \leq P/m + (1-1/m)p_j \leq (2-1/m)OPT$.

Question 107

For m=4, List Scheduling factor is:

A. 4/3

B. 7/4

C. 2

D. None of these

Answer: B. Explanation: 2 - 1/4 = 7/4.

Question 108

LPT sorts jobs in:

A. Nondecreasing processing time

B. Nonincreasing processing time

C. Increasing machine index

D. Random order only

Answer: B. Explanation: Longest Processing Time first sorts jobs largest to smallest.

Question 109

For m=3, classical LPT factor is:

A. 11/9

B. 5/3

C. 3/2

D. None of these

Answer: A. Explanation: 4/3 - 1/(3*3) = 4/3 - 1/9 = 11/9.

6 k-Center

Question 110

For centers C, d(s,C)=min distance to a center. The covering radius is:

A. Minimum over sites

B. Maximum over sites

C. Sum over sites

D. Maximum pairwise center distance

Answer: B. Explanation: The objective is the worst site's distance to its nearest center.

Question 111

Current nearest-center distances are 2,5,5,1. Farthest-first selects a site with distance:

A. 1

B. 2

C. 5

D. 13

Answer: C. Explanation: It chooses a farthest currently underserved site; either tied site with distance five works.

Question 112

Farthest-first for metric k-Center is a:

A. 1-approximation always

B. 2-approximation

C. k-approximation only for k=1

D. None of these

Answer: B. Explanation: Its metric packing argument gives a radius at most twice optimum.

Question 113

Any k+1 sites assigned to k optimal centers include two assigned to one center. Their distance is at most:

A. $r^*/2$

B. r^*

C. $2r^*$

D. $4r^*$

Answer: C. Explanation: Triangle inequality bounds their mutual distance by $r^* + r^*$.

Question 114

A radius test chooses pairwise-more-than-2r-separated centers and obtains k+1 centers. What follows?

A. A radius-r cover with k centers exists

B. No radius-r cover with at most k centers exists

C. OPT radius equals 2r exactly

D. None of these

Answer: B. Explanation: One radius-r center cannot cover two sites farther than 2r apart.

Question 115

Which k-Center claims are correct? *(Select all that apply.)*

A. It minimizes maximum distance to nearest selected center

B. Triangle inequality is used in factor-two proof

C. Farthest-first chooses the nearest uncovered site

D. Guarantee concerns maximum radius, not sum

Answer: A, B, D. Explanation: Farthest-first chooses the farthest site, and the objective is a maximum radius.

7 Set Cover

Question 116

U=1,2,3,4; S1=1,2, S2=2,3, S3=3,4. Which pair covers U?

A. S1 and S2

B. S1 and S3

C. S2 and S3

D. None of these

Answer: B. Explanation: S1 union S3 equals the entire universe.

Question 117

At a weighted greedy Set Cover step with uncovered set R, minimize:

A. $w(S_i)/|S_i \cap R|$

B. $|S_i \cap R|/w(S_i)$

C. $w(S_i)|S_i|$ regardless of R

D. None of these

Answer: A. Explanation: The score is cost per newly covered element; zero-new-coverage sets are ignored.

Question 118

Candidate sets cover 4, 4, 3, and 0 new elements, all with equal cost. Greedy best choice?

A. Only first

B. Only second

C. Either first or second

D. Fourth

Answer: C. Explanation: The two sets covering four new elements tie for best marginal coverage.

Question 119

What is H_5?

A. 137/60

B. 5/2

C. 1+ln 5 exactly

D. None of these

Answer: A. Explanation: 1+1/2+1/3+1/4+1/5 = 137/60.

Question 120

The standard greedy weighted Set Cover guarantee is:

A. H_n approximation with $H_n \leq 1 + \ln n$

B. 1/n approximation

C. Factor 2 for all n

D. Exact for every instance

Answer: A. Explanation: Greedy cost is at most H_n times OPT, with $H_n \leq 1 + \ln n$.

Question 121

Which Set Cover claims are correct? *(Select all that apply.)*

A. Every universe element must be covered

B. Previously covered elements do not count as new coverage

C. Greedy is always exactly optimal

D. Weighted greedy uses cost and new coverage

Answer: A, B, D. Explanation: The greedy guarantee is logarithmic in general, not exact on all instances.

8 Local Search and Bipartite Graphs

Question 122

No permitted one-step neighbor improves the current solution. What is established?

A. Local optimum under that neighborhood

B. Global optimum necessarily

C. Factor-2 guarantee necessarily

D. None of these

Answer: A. Explanation: The stopping condition establishes only neighborhood-local optimality.

Question 123

In a bipartite graph with sides L and R, every edge joins:

A. Two vertices in L

B. Two vertices in R

C. One vertex in L and one in R

D. Three vertices

Answer: C. Explanation: A bipartition permits edges only across the two sides.

Question 124

Minimum vertex-cover size of $K_{3,4}$ is:

A. 3

B. 4

C. 7

D. 12

Answer: A. Explanation: For complete bipartite $K_{a,b}$, the minimum cover size is $\min(a, b)$.

Question 125

A bipartite graph has maximum matching size 6. By König's theorem, minimum cover size is:

A. At most 3

B. Exactly 6

C. At least 12

D. Unrelated to matching

Answer: B. Explanation: König's theorem equates the two optimum cardinalities.

9 Metropolis and Hopfield

Question 126

At T=2, an uphill move has delta E=2. Acceptance probability is:

A. e^{-1}

B. e^{1}

C. 1/2

D. None of these

Answer: A. Explanation: The formula gives $e^{-\Delta E/T} = e^{-2/2} = e^{-1}$.

Question 127

For delta E=-3, standard Metropolis acceptance probability is:

A. $e^{-3/T}$

B. $e^{3/T}$

C. 1

D. 0

Answer: C. Explanation: Downhill moves are accepted with probability one.

Question 128

For fixed delta E>0, as T approaches zero from above, exp(-delta E/T) approaches:

A. 0

B. 1

C. delta E

D. Infinity

Answer: A. Explanation: The exponent tends to negative infinity.

Question 129

Which is geometric cooling?

A. $T_{t+1} = T_t - \alpha$

B. $T_{t+1} = \alpha T_t$, $0 < \alpha < 1$

C. $T_{t+1} = T_t + \alpha$

D. $T_{t+1} = T_t^2$

Answer: B. Explanation: Multiplying by a fixed factor below one gives geometric cooling.

Question 130

Which simulated annealing statements are correct? *(Select all that apply.)*

A. It may accept uphill moves

B. An uphill move with positive delta E at positive finite T has probability between zero and one

C. Cooling lowers acceptance probability for a fixed uphill move

D. It guarantees a global optimum in a fixed small number of steps

Answer: A, B, C. Explanation: The probability is exp(-delta E/T); no universal fixed small iteration guarantee exists.

Question 131

Standard binary Hopfield states are:

A. -1,+1

B. 0,1,2

C. All real values

D. Vertex permutations

Answer: A. Explanation: The common bipolar representation uses -1 and +1.

Question 132

Standard Hopfield energy descent commonly assumes:

A. Symmetric weights and zero self-weights

B. Antisymmetric weights and positive self-weights

C. All weights zero

D. None of these

Answer: A. Explanation: Symmetry and zero diagonal support the usual energy argument for asynchronous updates.

10 Randomized Selection and Sorting

Question 133

For sorted [4,8,11,17,23,29,35,41], the sixth smallest is:

A. 23

B. 29

C. 35

D. 41

Answer: B. Explanation: The sixth element is 29.

Question 134

Exactly ell distinct keys are smaller than pivot p. Its one-based rank is:

A. ell

B. ell+1

C. n-ell

D. n

Answer: B. Explanation: The pivot follows the ell smaller keys.

Question 135

Pivot rank is r and target rank k>r. Target rank in the right partition is:

A. k-r

B. k-r+1

C. k

D. r-k

Answer: A. Explanation: The right side excludes r positions up to and including the pivot.

Question 136

If Quickselect recurses on at most half the array, which recurrence gives linear time?

A. T(n)<=T(n/2)+cn

B. T(n)<=2T(n/2)+cn

C. T(n)=T(n-1)+cn

D. None of these

Answer: A. Explanation: The geometric sum cn+cn/2+cn/4+... is O(n).

Question 137

Why does T(n)=T(n-1)+cn yield Theta(n squared)?

A. Constant work at one level

B. Sum c(n+n-1+...+1)

C. Logarithmic depth with constant work

D. None of these

Answer: B. Explanation: The arithmetic series is n(n+1)/2.

Question 138

Ignoring rounding, what fraction of pivot ranks are guaranteed to leave both parts no larger than about 3n/4?

A. At least about 1/4

B. At least about 1/2

C. Exactly 3/4

D. None of these

Answer: B. Explanation: The middle half of ranks leaves neither side larger than roughly three quarters.

Question 139

If a good pivot occurs with probability at least 1/2, expected attempts until one occurs are at most:

A. 1/2

B. 1

C. 2

D. n/2

Answer: C. Explanation: A geometric waiting time has mean 1/p, at most 2 when p >= 1/2.

Question 140

If expected work before a good pivot is at most 2cn and the remaining size is at most 3n/4, which recurrence follows?

A. E[T(n)] <= 2cn + E[T(3n/4)]

B. E[T(n)] = 2E[T(n/2)] + cn

C. E[T(n)] <= E[T(n-1)]

D. None of these

Answer: A. Explanation: Expected attempts are at most two, and a good pivot leaves at most 3n/4 elements.

Question 141

Expanding E[T(n)] <= 2cn + E[T(3n/4)] gives which geometric bound?

A. $2cn \sum_i (3/4)^i = 8cn$

B. $cn \sum_i (1/2)^i = 2cn$

C. $2cn \sum_i (4/3)^i = O(n)$

D. None of these

Answer: A. Explanation: The series sums to 4, giving 2cn times 4 = 8cn.

Question 142

Randomized Quickselect has which complexity pair?

A. Worst Theta(n), expected Theta(n squared)

B. Worst Theta(n squared), expected Theta(n)

C. Both Theta(n log n)

D. None of these

Answer: B. Explanation: Expected linear time coexists with quadratic worst case from repeated extreme pivots.

Question 143

Balanced Quicksort recurrence T(n)=2T(n/2)+cn solves to:

A. Theta(n)

B. Theta(n log n)

C. Theta(n squared)

D. Theta(log n)

Answer: B. Explanation: The linear work per level over logarithmically many levels gives n log n.

11 Approximation Proofs and Traps

Question 144

ALG=24 and OPT >=16 for a minimization problem. What ratio bound follows?

A. ALG/OPT >= 3/2

B. ALG/OPT <= 3/2

C. OPT=16

D. None of these

Answer: B. Explanation: Since OPT is at least 16, ALG/OPT is at most 24/16.

Question 145

Which is valid for a rho-approximation to minimization?

A. ALG <= OPT/rho

B. OPT <= ALG <= rho OPT

C. ALG equals rho OPT always

D. None of these

Answer: B. Explanation: Feasibility gives ALG >= OPT, and the approximation guarantee gives ALG <= rho OPT.

Question 146

Which can be used in approximation proofs? *(Select all that apply.)*

A. Compare ALG to a lower bound on OPT

B. Prove a feasible solution's cost is at most a multiple of OPT

C. Show only that the algorithm terminates

D. Use matching or triangle inequality to relate ALG and OPT

Answer: A, B, D. Explanation: Termination alone does not prove solution quality; structural bounds relate the returned solution to OPT.

Question 147

What distinguishes a guarantee from a ratio observed on one instance?

A. A guarantee is proved for all valid instances; an observed ratio concerns one instance

B. They are always identical

C. One good instance proves the worst-case guarantee

D. None of these

Answer: A. Explanation: A single instance cannot establish a universal bound.

Question 148

ALG=40 and OPT=32. The minimization approximation ratio on this instance is:

A. 4/5

B. 5/4

C. 8

D. None of these

Answer: B. Explanation: ALG/OPT=40/32=5/4.

Question 149

A factor-2 minimization approximation means:

A. Output cost is at most twice OPT, not necessarily exactly twice OPT

B. Output is at least twice OPT

C. Output is always optimal

D. None of these

Answer: A. Explanation: The factor is an upper bound on the ratio, not an equality.

9

Question 150

Which pairing is correct?

A. Metric TSP MST preorder factor 2; metric k-Center farthest-first factor 2

B. Greedy Set Cover exact; List Scheduling factor 1

C. Vertex Cover edge-picking factor 1; LPT factor 2 for every m

D. None of these

Answer: A. Explanation: Both named metric algorithms have standard factor-2 guarantees.

Last-Minute Formula and Logic Checklist

Last-Minute Formula and Logic Checklist

Concept	Formula / Guarantee
Complexity classes	$\mathbf{P} \subseteq \mathbf{NP} \subseteq \mathbf{PSPACE}$
Vertex Cover (edge-picking)	$\lvert C\rvert \leq 2\lvert C^*\rvert$
Pricing lower bound	$\sum_{e \in E} p_e \leq OPT$
Metric TSP (preorder traversal)	$w(H) \leq 2w(T) \leq 2OPT$
List Scheduling	$T \leq \left(2 - \frac{1}{m}\right) OPT$
LPT Scheduling	$T \leq \left(\frac{4}{3} - \frac{1}{3m}\right) OPT$
Metric k-Center	$r(C) \leq 2r^*$
Greedy Set Cover	$ALG \leq H_n OPT \leq (1 + \ln n) OPT$
Metropolis (uphill move)	$p = e^{-\Delta E/T}, \quad \Delta E > 0$
Quickselect (expected time)	$\Theta(n)$
Randomized Quicksort (expected time)	$\Theta(n \log n)$
Quickselect and Quicksort (worst case)	$\Theta(n^2)$

Common traps: A reduction from a known hard problem A to B transfers hardness to B; for minimization, compare algorithm cost with OPT; lower bounds on OPT help upper-bound the ratio; a local optimum need not be global; randomization improves expected performance but does not rule out a bad run.